



University of San Carlos | Department of
COMPUTER ENGINEERING

CpE 3104 – Microprocessors

8086 μ P Bus Buffering and Latching

Unit 5: MEMORY INTERFACING

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Bus

■ Address bus

- It provides the memory and I/O with the *memory address* and *I/O port number*.

■ Data bus

- This bus *transfers data* between the *microprocessor* and the *memory* and *I/O* in the system.

■ Control bus

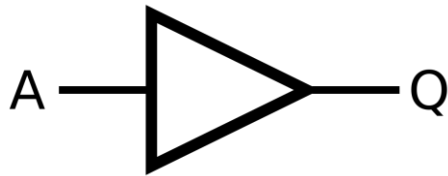
- It provides *control signals* to the *memory* and *I/O*.

Buffer (Digital Buffer)

- It is a special electronic circuit that is used to isolate the input from the output, preventing the impedance of one circuit to alter the impedance of another..
- It has only a single input and a single output behavior that is the opposite of a NOT gate.
- It simply passes its input, unchanged, to its output.

Buffer (Digital Buffer): Important Points

- In a **Boolean logic simulator**, a **buffer** is mainly used to increase propagation delay.
- In a **real-world circuit**, a **buffer** can be used to amplify a signal if its current is too weak.



TRUTH TABLE OF A BUFFER

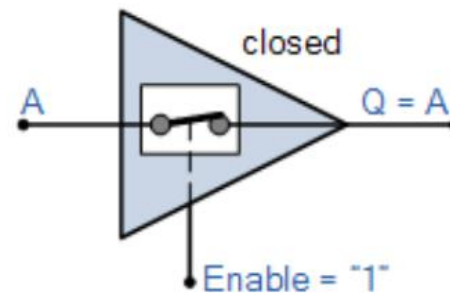
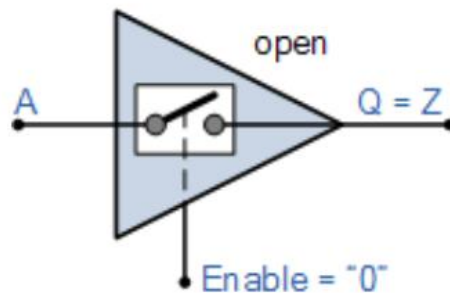
Input	Output
false (0)	false (0)
true (1)	true (1)

Latch

- It is a form of sequential logic circuit that, for a simple latch, has two stable logic states that maintains its states indefinitely until an input pulse called a **trigger** is received.
- If a trigger is received, the latch outputs change states according to defined rules, and remain in those states until another trigger is received.

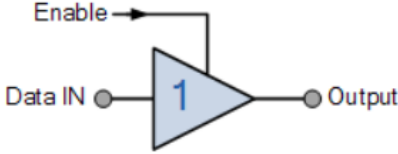
What is a Tri-State or Three-State Device?

- Refers to any digital device wherein its output has three electrical states:
 1. *High output state (1)*
 2. *Low output state (0)*
 3. *High-impedance / Hi-Z state (disconnected/off)*
 - A state in which the output is disconnected, leaving the signal open, to be driven by another device (or to be pulled up or down by a resistor provided to prevent an undefined state).

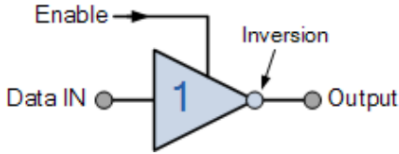


Tri-State Buffer

Active HIGH

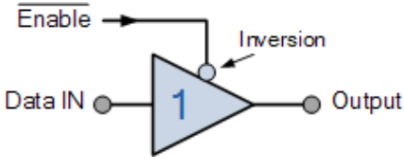
Symbol	Truth Table		
 Tri-state Buffer	Enable	IN	OUT
	0	0	Hi-Z
	0	1	Hi-Z
	1	0	0
	1	1	1
Read as Output = Input if Enable is equal to "1"			

Active HIGH Inverting

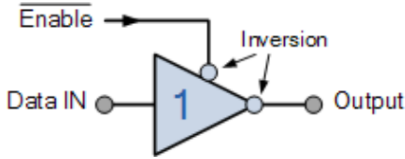
Symbol	Truth Table		
 Inverting Tri-state Buffer	Enable	IN	OUT
	0	0	Hi-Z
	0	1	Hi-Z
	1	0	1
	1	1	0
Read as Output = Inverted Input if Enable equals "1"			

Tri-State Buffer

Active LOW

Symbol	Truth Table		
 Tri-state Buffer	Enable	IN	OUT
	0	0	0
	0	1	1
	1	0	Hi-Z
	1	1	Hi-Z
Read as Output = Input if Enable is NOT equal to "1"			

Active LOW Inverting

Symbol	Truth Table		
 Inverting Tri-state Buffer	Enable	IN	OUT
	0	0	1
	0	1	0
	1	0	Hi-Z
	1	1	Hi-Z
Read as Output = Inverted Input if Enable is NOT equal to "1"			

8086 Microprocessor Multiplexed Buses

$AD_0 - AD_{15}$ (Address/Data Signals)

$A_{16}/S_3 - A_{19}/S_6$ (Address/Status Signals)

$\overline{BHE}/S_7$ (Bus High Enable/Status Signal)

Why Not Leave the Buses Multiplexed?

- Memory and I/O **require** that the address remains valid and stable throughout a **read or write cycle**.
- If the buses are multiplexed, the address changes at the memory and I/O, which causes them to read or write data in the wrong locations.

Demultiplexing Buses

- It is a **process** of **separating** the shared bus lines or bus signals.

- It may use **latches**, **buffers** or **transceivers** for the **demultiplexing process**.
 - **Latches** are used for **address bus signals**.
 - **Buffers or transceivers** are used for **data bus signals**.

Latch: 74LS373

- It is an IC that contains 8 binary digital circuits called latches.
- This IC contains **octal D-type transparent latches** with **3-state outputs** for bus organized system applications.
- While the **enable (G/LE)** is **HIGH**, the **Q outputs** will follow the **data (D) inputs** and when the **enable** is taken **LOW**, the **output** will be latched at the level of the data that was set up.

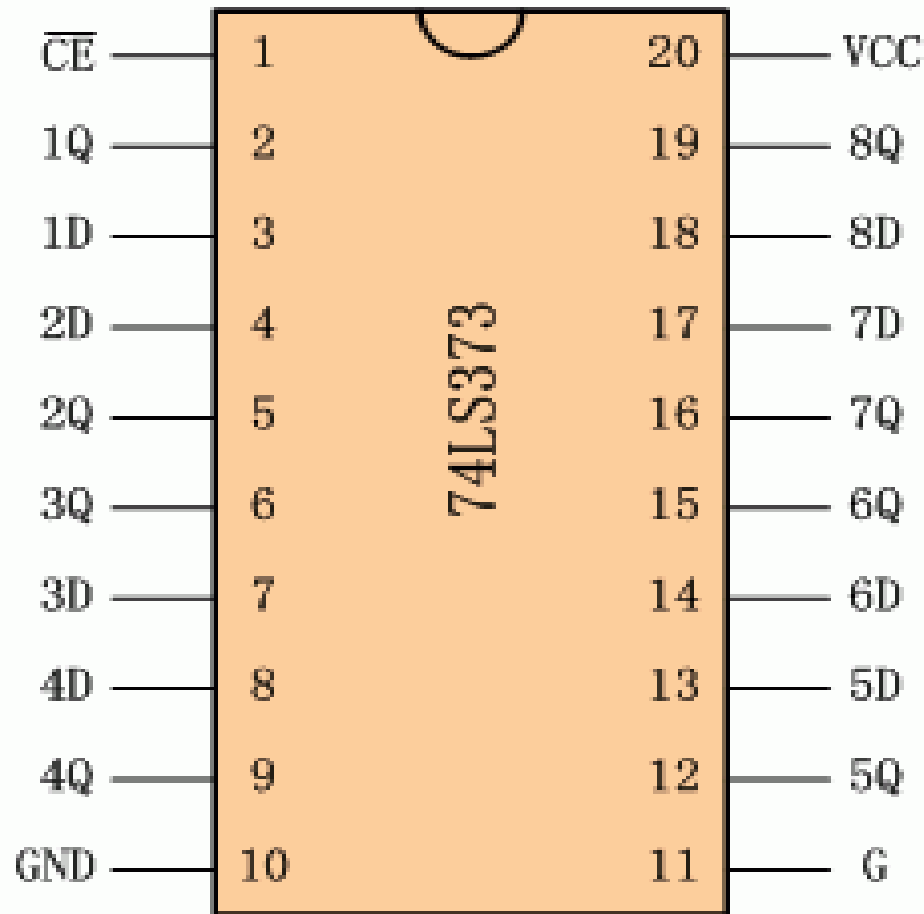
Latch: 74LS373

- **Data appears on the bus** when the Output Enable/Output Control ($\overline{OE}/\overline{OC}/\overline{CE}$) is **LOW**.
- When $\overline{OE}/\overline{OC}/\overline{CE}$ is **HIGH**, the **bus output** is in the **high impedance state**.
- The **latch** can **hold onto the data at its inputs before transmitting the data to its outputs**.

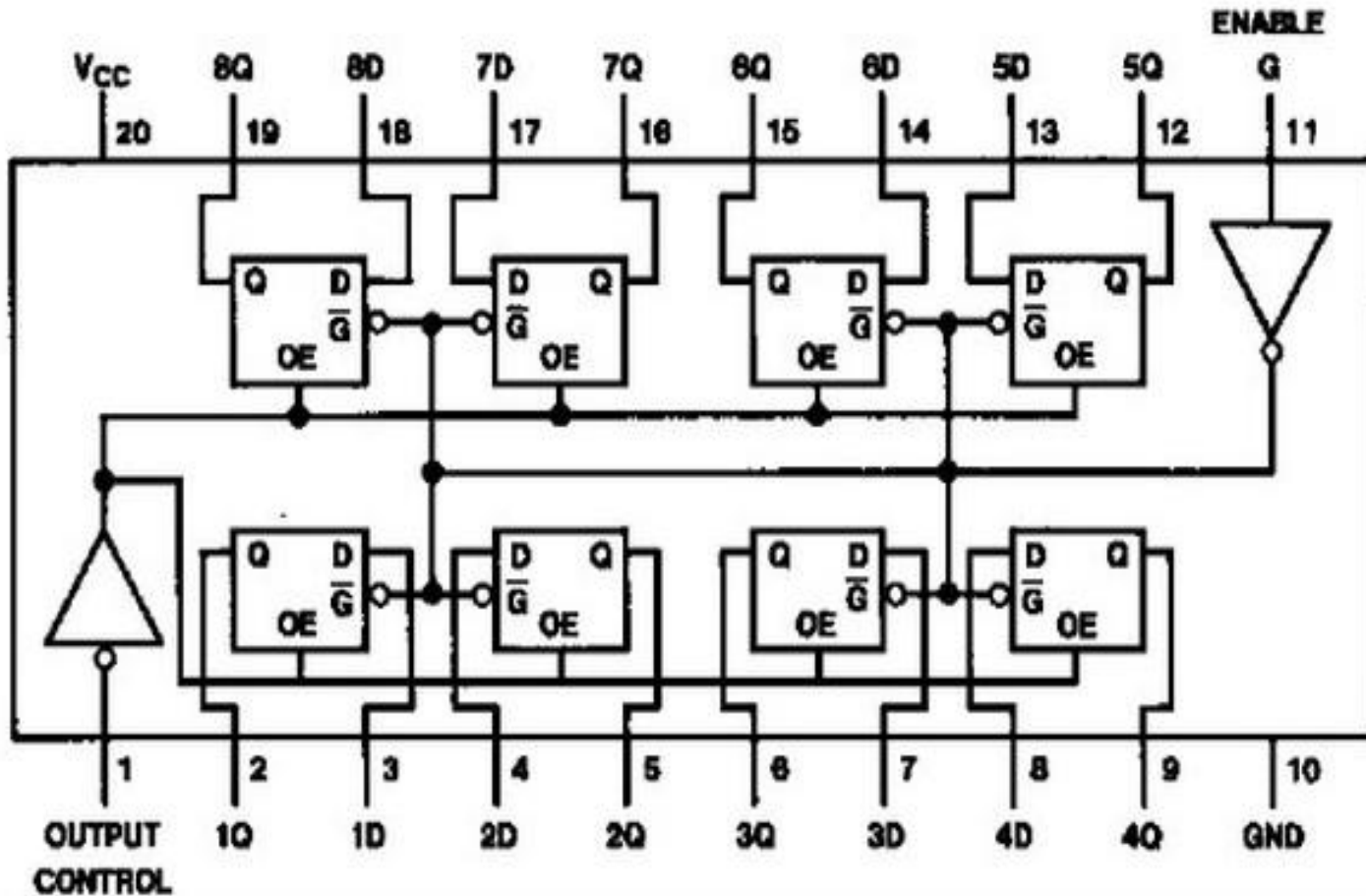
Latch: 74LS373 Function Table

OUTPUT CONTROL (Active low)	ENABLE (Active high)	DATA (D)	OUTPUT (Q)
L	H	H	H (High Level, Steady State)
L	H	L	L (Low Level, Steady State)
L	L	X	Q_0 (The level of the output before the Steady-State input conditions were established; no change)
H	X	X	Z (High-Impedance State)

Latch: 74LS373 Pin Configuration



74LS373 Internal Connection Diagram



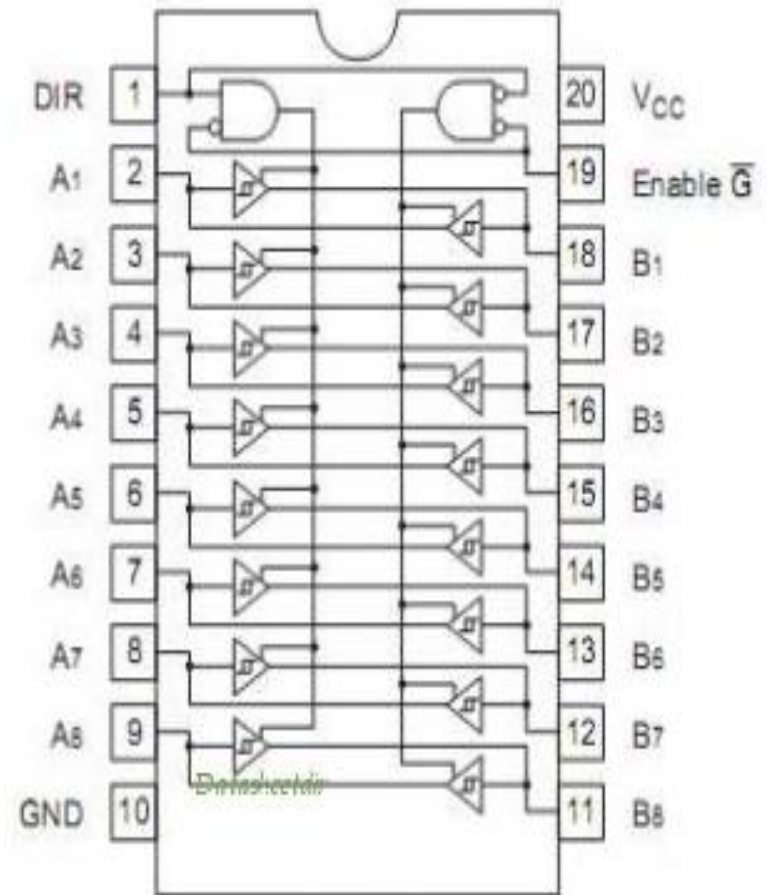
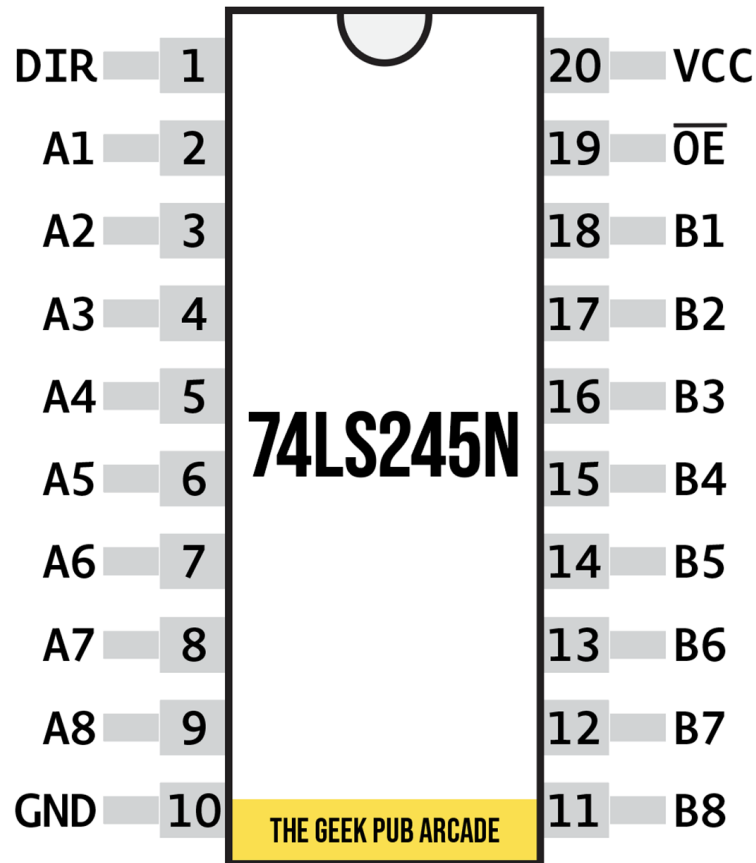
Octal Transceiver: 74LS245

- It is a **tri-state OCTAL BUS TRANSCEIVER (bidirectional buffer)**.
- It is designed for **asynchronous two-way communication between data buses**.
- **Allows data transmission from the A bus to the B bus or from B bus to the A bus, depending on the logic level at the direction-control (DIR) input.**
- The **output enable ($\overline{OE}/\overline{G}$) input can disable the device so that the buses are effectively isolated.**

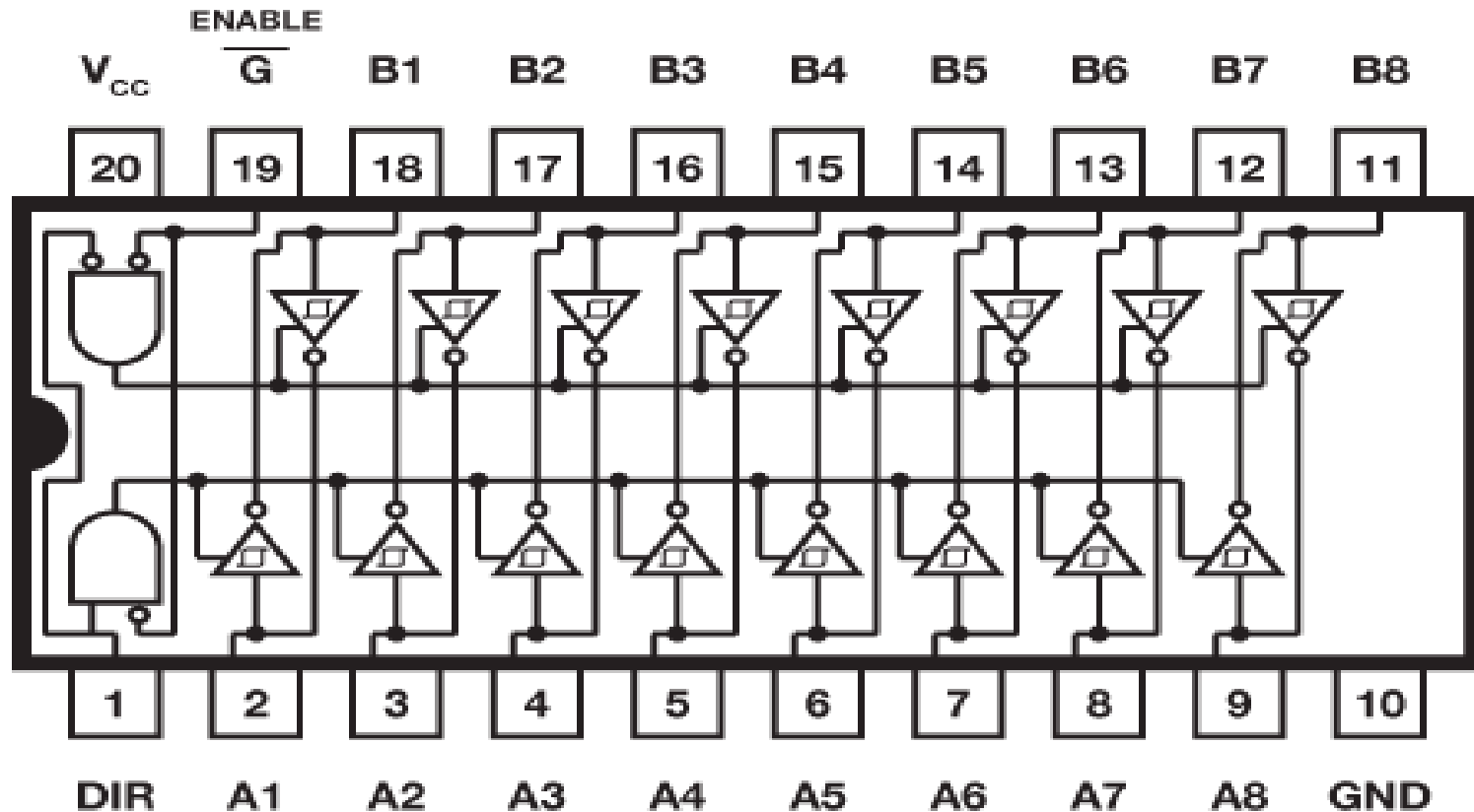
Octal Transceiver: 74LS245 Function Table

INPUTS		OPERATIONS
$\overline{\text{OE}}$	DIR	
L	L	B data to A bus
L	H	A data to B bus
H	X	Isolation

Octal Transceiver: 74LS245 Pin Configuration



74LS245 Internal Connection Diagram



Problem

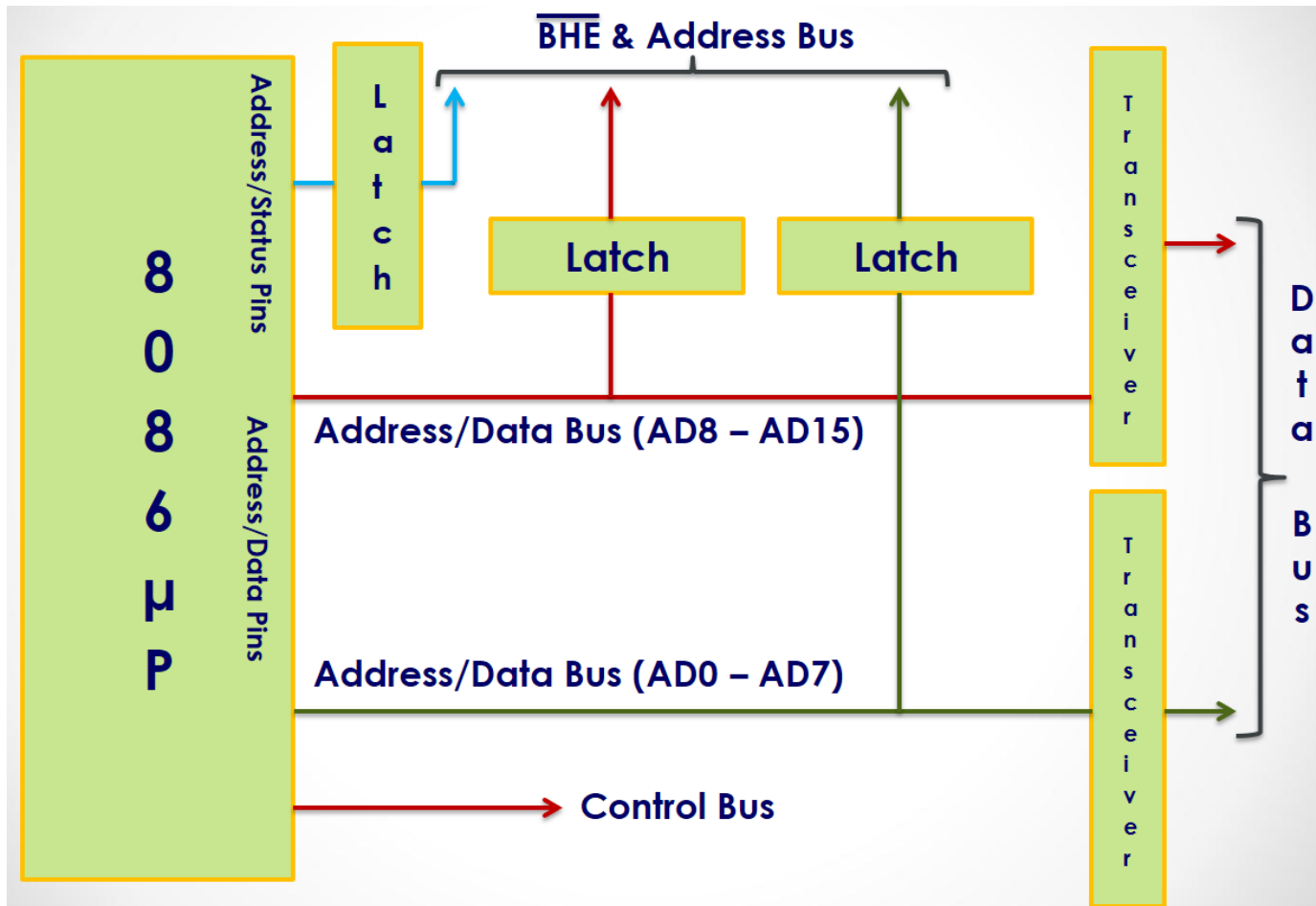
Given:

The 8086 μP has **several multiplexed buses**. The **multiplexed buses must be demultiplexed** before the *8086 μP is used with memory and I/O interfaces* using latches and buffers.

Required Task:

Draw (using the required pencil and paper) the **complete circuit diagram of demultiplexing the 8086 μP multiplexed buses**.

Block Diagram: Demultiplexing the 8086 μ P Buses





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End of Lecture